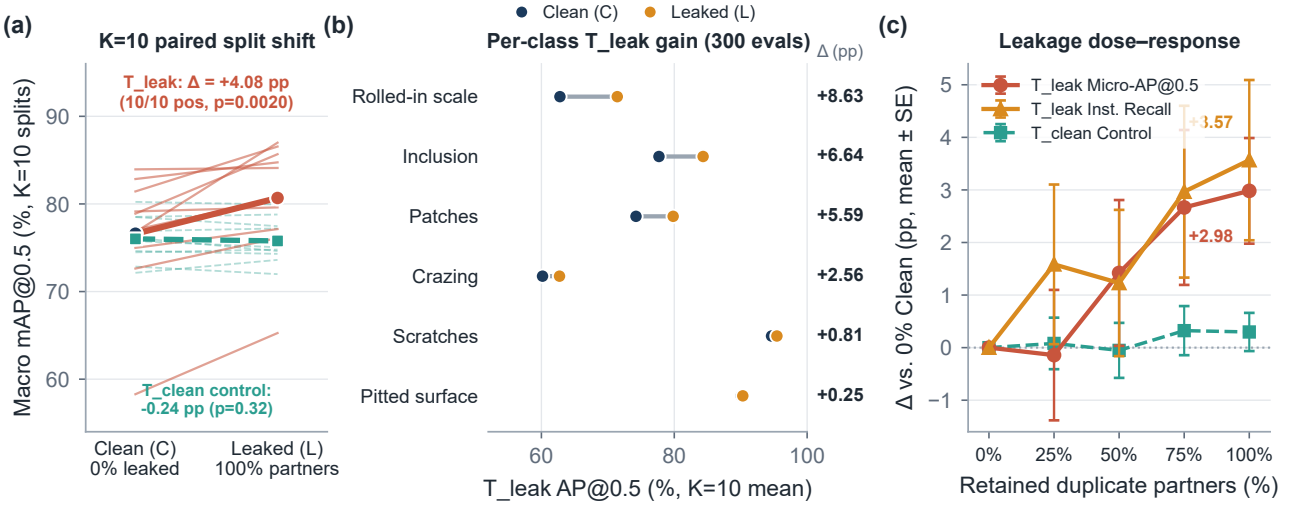

(a)
K=10 paired split shift
T_leak: Δ = +4.08 pp
(10/10 pos, p=0.0020)
T_clean control:
-0.24 pp (p=0.32)
Macro mAP@0.5 (%, K=10 splits)
90
80
70
60
Clean (C)
0% leaked
Leaked (L)
100% partners
(b)
Clean (C)
Leaked (L)
Per-class T_leak gain (300 evals)
Δ (pp)
Rolled-in scale
Inclusion
Patches
Crazing
Scratches
Pitted surface
+8.63
+6.64
+5.59
+2.56
+0.81
+0.25
60
80
100
T_leak AP@0.5 (%, K=10 mean)
(c)
Leakage dose–response
T_leak Micro-AP@0.5
T_leak Inst. Recall
T_clean Control
+3.57
+2.98
Δ vs. 0% Clean (pp, mean ± SE)
5
4
3
2
1
0
−1
0%
25%
50%
75%
100%
Retained duplicate partners (%)